Zachary Gray

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Education

Bachelor of Science in Electrical Engineering

California Polytechnic State University, San Luis Obispo, CA

- Graduating as Undergraduate June 2026
- 3.7 GPA
- Relevant Coursework: Advanced Photonics Engineering, Lasers, RF/Microwave Design

Master of Science in Electrical Engineering

California Polytechnic State University, San Luis Obispo, CA

- Blended Master of Science in Electrical Engineering (4+1) Expected June 2027
- Concentrating in Photonics and RF

Technical Skills

RF & Test Equipment: Vector Network Analyzer (VNA), Spectrum Analyzer, Oscilloscope, PlutoSDR
Programming: Python, MATLAB, C, SystemVerilog, Git
RF & Simulation Tools: GNU Radio, ADS, LTSpice, Keysight Photonics Designer, Code V (coursework)
Embedded Systems: STM32, Arduino, SPI, I2C, UART
Concepts: Signal Integrity, Eye Diagram Analysis, Mixed Signal Systems, S-Parameters, Modulation Theory, Behavioral Modeling

Projects

Bachelor's Thesis: mm-wave Based Measurement of Air Humidity

- Designed and tested 24 GHz SFCW radar system to estimate atmospheric humidity across varying environmental conditions
- Built behavioral signal processing pipeline in Python to model radar return characteristics and correlate against reference humidity data; achieved $R^2 = 0.88$
- Automated end-to-end data acquisition, processing, and analysis using Python scripting

LPI Comms: Frequency Hopping Spread Spectrum Link (2.4 GHz)

- Designed receiver for 4 QAM frequency-switching communication system in Wi-Fi band
- Implemented modulation/demodulation using PlutoSDR and GNU Radio with Python
- Performed eye diagram analysis and analyzed constellation diagrams to characterize timing jitter and inter-symbol interference (ISI), diagnosing and resolving link timing errors

Antenna Measurement System:

- Contributed to development of automated antenna measurement system
- Designed microstrip band pass filter (<2 dB insertion loss in passband)

Experience

Conex Electro, Electrical Engineering Intern, Bellingham, WA

- Summer 2025
  - Conducted PCB inspection and electrical validation of barometer modules
  - Used oscilloscopes and multimeters to verify signal integrity and sensor operation

Cal Poly Electrical Engineering Dept, Instructional Student Assistant, San Luis Obispo, CA

- February 2025 – June 2025
  - Graded RF coursework and lab reports

Languages

Spanish – Advanced conversational proficiency